Lane Brief 2A — Feature Specification, Firewall Tests and Baselines

Field	Detail
Owner	Shravani Math
Role	Feature-specification and verification owner for Lane 2
Reassigned from	Arshiya Mahammad (resigned)
Branch	`feat/features-harness` — you commit; Sathvik owns and merges the branch
Blocks	A, B, C — see the shared schedule
Date	14 November 2026
Companion brief	`docs/team/02b_sathvik_features_folds.md` — see `satv.pdf`

1. Your mission in one paragraph

Define, in writing, exactly what a legal predictor is for this project, then prove it with tests that are designed to fail. Sathvik builds the feature table and the fold file; you own the specification they must satisfy, the independent leakage tests that police it, and the first honest baselines. You are not the table builder. Your value is that a second person, working from the label side, can say "this column is safe, and here is the test that shows it" without having written the builder. If the table leaks, the project has no honest result — that is the job you are protecting.

2. Where we actually stand

Verified on `feat/india-scope-docs` at `4556011`, clean and synced with origin.

Item	State
USA clean panel	`data/clean/USA_panel.csv`, 267 quarters × 27 columns (key + 26 series), 1960Q1–2026Q3
USA label	`labels-v3` frozen, 147 quarters, 1990Q1–2026Q3, hash `2379fd53…19daa`
Label-ready growth	`data/clean/USA_growth.parquet`, eight series
NBER validation	`data/clean/USA_validation.parquet` — never a feature
Feature table	**does not exist**
Fold file	**does not exist**
Harness	**does not exist**
Test suite	53 passed, 1 skipped
`make check`	33 ok, 3 warnings (optional model packages), 0

Item	State
	failures
India	1/8 downloaded (`ACT_IP` only); Label-B-only under O-2
Lane 3	blocked: no official run, `config/models.yaml` unapproved

What this means for you: there is nothing half-built to inherit. You start from a clean panel and a frozen label. That is the best possible position for writing a spec.

Why Lane 2 was split. Arshiya resigned before delivering any of Lane 2. Rather than have one person build and self-certify, the lane is split: Sathvik builds, you specify and verify. This is deliberate separation of duties, not a demotion of the work.

3. The whole remaining work, and which part is yours

This is the full path from today to the final go/no-go. Your name marks what you own; “support” means you review or unblock rather than build.

Workstream	State	Owner
JPN window closeout: 144 vs 147 quarters, label-doc hash drift	open	**You**
Lane 2A — feature spec, firewall tests, baselines	not started	**You**
Lane 2B — table, folds, harness, Makefile	not started	Sathvik
Repo reconciliation: merge `origin/master`, dependency install, doc drift	pending	Sathvik
Lane 3 — config freeze, intake gate, official run, C-13 once, results docs	blocked on 2+3+4	Palchhin (support: you)
Lane 4 closeout — India 1/8 recorded, scope audit, pitch	complete	Sathvik
Day 6 merge in fixed order	pending	each owner; Sathvik coordinates
Day 7 combined clean clone, hash table, go/no-go	pending	Sathvik runs; **you cross-check**; Palchhin results docs
Dashboard refresh from official artifacts	pending	Sathvik
Packaging / tag / mirror	blocked, needs explicit approval	Sathvik

You touch the JPN closeout, Lane 2A, and the cross-check half of Day 7. Everything else is either Sathvik’s build or Palchhin’s run. Do not absorb Lane 3 to make the story complete.

4. What you own

Block A — Saturday afternoon (2–3 h): the specification

Write `docs/FEATURES.md`. Sections 1–4 are yours; section 5 (dictionary) is Block C.

1. **Purpose and window.** USA predictors, 1990Q1–2026Q3, one row per quarter. State that the table is unscaled and that scaling happens inside the fold.
2. **Column inventory.** All 26 panel columns plus the eight label-ready growth series. For each: source series ID, native frequency, transform (level, Δlog, quarterly mean, etc.), availability lag under C-05, missing-data rule, and one line on why it is safe.
3. **Allowed raw BEAI copies.** The eight BEAI-eligible core series may appear **only** as separate, versioned predictor copies with their own transform and lag. Name them explicitly (a `beai_copy_` prefix or equivalent), and state that no fitted object, scaler or factor loading is shared with Pipeline A. This is the hardest paragraph in the document; get it right.
4. **Exclusion blacklist.** Write down every column class that is forbidden and why: `factor`, `level`, `R`, `n_available`, `deterioration_share`, turning-point flags, `USA.NBER.Q`, country ID, country-size levels, and anything derived from future observations.
5. **2026 extension audit.** What changed when the window moved from 2025Q4 to 2026Q3, and why it does not create a new leak.

A useful sanity check: every row of the spec should let a stranger rebuild that column without asking you a question.

Block B — Sunday (4–5 h): the tests

Write `tests/test_features.py` against Sathvik's builder. The file is yours; do not accept edits that weaken it.

Test	What it must prove
Schema	147 rows, `quarter` key, contiguous 1990Q1–2026Q3, no duplicate keys
Blacklist	None of the forbidden columns or prefixes appear
Unscaled	Values are not standardised; no zero-mean/unit-variance signature per column
No future	Deleting all rows after `t` does not change any feature value at `t`
Independence	No BEAI-derived object or shared scaler is used in Pipeline B
Mutation	A deliberately shuffled or leaking copy of the table **fails** the suite

The mutation test is the important one. A test that only passes on the correct table proves nothing; a test that fails on a known-bad table proves the firewall.

Block C — Monday (3 h): dictionary and baselines

- **Feature dictionary** (`docs/FEATURES.md` §5): units, interpretation, expected sign, known coverage gaps, and a short "what I learned about leakage" section.
- **Baselines** (`scripts/07_baselines.py`) on Sathvik's frozen folds: majority class, term-spread probit/logit, plain logistic. Report pooled PR-AUC and episode recall. No tuning, no feature selection on the test block, no threshold search.
- **Hash proof:** re-run the USA label hash check and record `2379fd53…19daa` unchanged.

5. Deliverables

1. `docs/FEATURES.md` — spec §1–4, dictionary §5.
2. `tests/test_features.py` — all six tests, including the mutation test.
3. `scripts/07_baselines.py` — three baselines on the frozen folds.
4. A short baseline results note (`outputs/results/baselines.md` or equivalent, or a section in `docs/FEATURES.md`) with PR-AUC, recall and the seed.
5. Comment on Sathvik's PR with the test evidence and the label hash.
6. Closeout note on the JPN window (see §7).

6. Done when — minimal gates

These five are non-negotiable. Everything else can slip.

1. `make features` yields 147 rows, 1990Q1–2026Q3, unscaled, keyed on `quarter`, with zero blacklisted columns — **and you agree it is correct**.
2. `make folds` is deterministic; no test row ever appears in training; 1Q/2Q/4Q exist with 2Q primary.
3. The mutation test fails on a deliberately leaking table.
4. USA label hash is still `2379fd53…19daa`; estimator, BBQ, diffusion gate and the eight-indicator structure are untouched.
5. One USA country × horizon reproduces from a recorded seed.

Deferred, not required: extra predictors, missingness plots, fold diagrams, fourth baseline, cross-country feature panels. Add them only if Blocks A–C finish early.

7. Carryover: the Japan window (small, separate)

`labels-v1-JPN` covers 144 quarters (1990Q1–2025Q4). `config/scope.yaml`, D-R02 and D-R31 say 147 (1990Q1–2026Q3). There is no decision-log entry for 144, and `docs/LABEL_VALIDATION_JPN.md` also records the wrong metadata hash.

Pick one and write it down before any cross-country feature work:

- **Option A (cheap):** record 144 as a documented exception with a decision-log entry, and correct the metadata hash in the validation note.
- **Option B (correct, slower):** rebuild the Japan label as `labels-v2-JPN` on the full 147-quarter window, leaving `labels-v1-JPN` frozen and untouched.

Recommendation: **Option A now**, Option B only if the official run leaves time. The USA-first headline is not blocked either way. Do not silently let the docs and the label disagree.

8. How your work is judged

- **Feature gate:** no BEAI-derived output, NBER series, future data, country ID or country-size level. Allowed raw copies are documented and provably independent.
- **Test independence:** you did not write the builder, and a known-bad table fails your suite.
- **No collateral change:** the label stays byte-identical.
- **Honesty:** a predictor that fails coverage is documented and excluded, never imputed.

9. Hand-offs

ID	To	When	What must be true
—	Sathvik	End of Block A	Spec §1–4 is written; he can build without asking you
—	Sathvik	End of Block B	Tests run green on his table; mutation test fails on a bad one
H2/H4 (adapted)	Palchhin	End of Block B	Draft folds + frozen table; he can run a first real pass
H4	Palchhin	End of Block C	Baselines + dictionary; no tuning after this point
Day 7	Sathvik	After official run	You independently reproduce the clean-clone checks

10. DOs

- DO read C-01, C-05–C-08, C-11 and C-12 before writing a single spec line.
- DO write the spec before Sathvik writes the table. He is blocked on you.
- DO keep the table unscaled and keep the test block untouched.
- DO test the firewall with a deliberately leaking table and a shuffled pipeline.
- DO ask Palchhin what the folds must expose (keys, horizons, inner folds) before you freeze.
- DO commit small, run the suite after each change, and write a hand-off the receiver can follow.

11. DON'Ts

- DON'T use the BEAI factor, R, level, diffusion, turning-point flag or NBER column as a feature.
- DON'T use future data, full-sample scaling, or full-sample feature selection.
- DON'T interpolate, forward-fill or impute a partial quarter.
- DON'T include a country ID or a raw country-size level.
- DON'T hand-edit `dataset.parquet` or `folds.json`; regenerate them.
- DON'T weaken a failing leak test to make the table pass.
- DON'T change the label or the frozen rules to make the table easier.
- DON'T overclaim: a good-looking baseline score is not evidence until the leak tests pass.

12. Risks and how to cut

Risk	Trigger	Cut
Spec becomes a research project	Still writing §2 on Sunday morning	Cap transforms to C-05 defaults; document the rest as open questions
Test file fights the builder	Two days of red tests	Freeze on schema + no-future + mutation; move independence to a review note
Baselines drag	No folds until late Sunday	Ship majority + logistic only; note probit as deferred
Japan window expands	Option B chosen	Defer to after the official run; USA-first is unaffected

The critical path is the spec. If Block A slips, Sathvik's Block B slips, and Palchhin's run slips. Send the spec in whatever state it reaches by Saturday night.

13. Where to look

Need	File
Frozen rules	`docs/BUILD_CONTRACT.md` (C-01, C-05–C-08, C-11, C-12)
Decisions and rationale	`docs/DECISIONS.md` (D-R05, D-R06, D-R08, D-R12–D-R18, D-R22–D-R31)
USA clean panel	`data/clean/USA_panel.csv`
Label-ready growth	`data/clean/USA_growth.parquet`
NBER validation only	`data/clean/USA_validation.parquet`
Scope and sources	`config/scope.yaml`, `data/scope.csv`
Harmoniser	`scripts/harmonize.py`
Your Lane 1 outputs	`docs/LABEL_VALIDATION_JPN.md`, `docs/HANDOFF_JPN_DEU.md`
Original Lane 2 brief	`docs/team/02_arshiya_features_harness.md`
Sathvik's build brief	`docs/team/02b_sathvik_features_folds.md`

Bottom line: a small, clean, tested table with honest folds beats a large table with one leak. You are the person who decides what "clean" means and proves it. Write the spec first, hand it over Saturday night, and let the tests do the arguing.